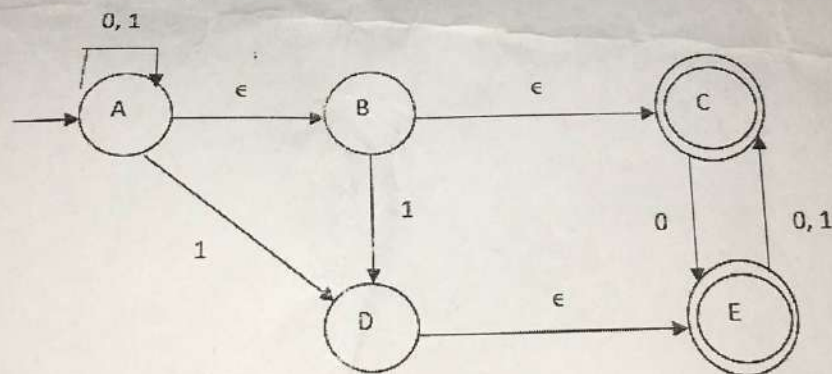


Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define the terms alphabet, string, kleene closure, positive closure, Language, and Regular Expression with examples. For language $L = \{w \in \{0, 1\}^* : w \text{ has even number of 0 followed by odd number of 1}\}$, find equivalent Strings and Regular Expression. 4+4
- b) Design a DFA that accepts the language given by $L = \{w \in \{0, 1\}^* : w \text{ starts with 01 having even length. Hence test your design for 0101110 and 0101}\}$. 7
2. a) Write about closure properties of regular language with suitable examples. 8
- b) Construct a DFA equivalent to following NFA. 7



3. a) "For every regular expression there is a finite automaton and for every finite automaton there is regular expression." Justify this statement with examples. 7
- b) Explain Chomsky Normal Form (CNF) and Greibach Normal Form (GNF). Convert following CFG to GNF 8
 $S \rightarrow XA \mid BB$
 $B \rightarrow b \mid SB$
 $X \rightarrow b$
 $A \rightarrow a$

4. a) Explain Pushdown Automata with its configuration. Design a PDA that accepts $L = \{wcw^R : w \in \{0,1\}^*\}$ and test your design for 0101C1010 and 011C101. 2+6

b) "For every CFG there is an equivalent Push Down Automata". Justify this statement with an example. 7

OR

State the pumping lemma for context free language. Show that $L = \{a^n b^n c^n : n \geq 1\}$ is not a context free language by using pumping lemma.

5. a) Explain Turing machine along with its formal definition. Design a Turing machine accepting $\{a^n b^n c^n : n \geq 1\}$. Test your design for aabbcc and aabbc. 2+6

OR

Define instantaneous description of Turing machine. How does Turing machine compute a function? Explain with example.

b) Explain encoding of Turing machine and a string with an example. 7

6. a) What is Computational Complexity Theory in Theory of Computation? Explain asymptotic analysis with different asymptotic notations. 8

b) Explain different types of complexity classes. What are tractable and intractable problems? Differentiate Computational, Decision and Optimization Problems with examples. 7

7. Write short notes on: (Any two) 2×5

a) Pumping lemma for Regular Language

b) Church Turing Thesis and Halting Problem

c) Cook's Theorem on Reducibility